VisionCare AI: An Explainable Dual-Neural Consensus and Attention U-Net Architecture for Multi-Class Retinal Disease Diagnosis and Clinical PACS Telemetry

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Abstract*—Optical Coherence Tomography (OCT) is the clinical gold standard, non-invasive imaging modality for evaluating cross-sectional retinal microarchitecture and diagnosing severe maculopathies, including Choroidal Neovascularization (CNV), Diabetic Macular Edema (DME), and Drusen. However, conventional automated diagnostic algorithms often suffer from high compute latency, lack of visual interpretability, and single-model vulnerability to out-of-distribution noise. In this paper, we present VisionCare, an end-to-end clinical-grade artificial intelligence diagnostic and Picture Archiving and Communication System (PACS) telemetry platform. The diagnostic engine introduces a weighted dual-architecture neural consensus combining an inverted-residual EfficientNet-B0 and a lightweight MobileNet-V3 backbone with soft probability voting (weights 0.70 and 0.30, respectively). Epistemic diagnostic uncertainty is quantified in real time through Shannon entropy over output probability distributions. Semantic lesion localization is driven by a MONAI-based Attention U-Net and DeepLabV3+ network with spatial gating to segment subretinal fluid, intraretinal fluid, and pigment epithelial detachments. Furthermore, a morphological foveal contour profile algorithm extracts real-time central subfield thickness across the Early Treatment Diabetic Retinopathy Study (ETDRS) 9-zone macular grid. Validated on a multi-class benchmark partition of the Kermany retinal OCT dataset encompassing 11 clinical classifications, our consensus model achieves an overall diagnostic accuracy of 95.01%, macro precision of 95.04%, recall of 95.01%, macro F1-score of 94.97%, and an Area Under the ROC Curve (ROC-AUC) of 0.9984 with sub-450 millisecond CPU inference latency. The system integrates explainable Grad-CAM saliency heatmaps, interactive 3D macular elevation mapping, digital micrometer calipers calibrated at 3.9 micrometers per pixel, an automated bilingual (English and Tamil) clinical reporting pipeline, and a Retrieval-Augmented Generation (RAG) ophthalmic diagnostic assistant.

Keywords*—*Optical Coherence Tomography (OCT), Dual Neural Consensus, EfficientNet-B0, MobileNet-V3, Attention U-Net, Shannon Entropy Uncertainty, Explainable AI (Grad-CAM), Clinical PACS, Bilingual Automated Reporting.

I. INTRODUCTION

Retinal degenerative disorders and microvascular complications—most notably Age-Related Macular Degeneration (AMD), Diabetic Macular Edema (DME), and Choroidal Neovascularization (CNV)—represent the leading causes of irreversible vision impairment and legal blindness globally. According to World Health Organization (WHO) epidemiological studies, Diabetic Retinopathy and Macular Degeneration affect over 300 million individuals worldwide, with disproportionate morbidity in developing regions where specialized retina care is scarce, expensive, or delayed. Spectral-Domain Optical Coherence Tomography (SD-OCT) provides micron-resolution cross-sectional B-scans of the retina, enabling clinicians to visualize neurosensory cellular laminations ranging from the Internal Limiting Membrane (ILM) down to the Retinal Pigment Epithelium (RPE) and choroid layer.

Despite the clinical efficacy of SD-OCT, clinical image volume is expanding exponentially, resulting in severe clinical fatigue and diagnostic delays. Traditional computerized screening approaches present four fundamental bottlenecks: 1) Model Brittleness and Overconfidence: Conventional monolithic Deep Convolutional Neural Networks (DCNNs) produce overconfident misclassifications on borderline pathologies (e.g., distinguishing early Drusen from early CNV) without quantifying epistemic uncertainty. 2) Opacity of Interpretability: Black-box models fail to furnish verifiable visual evidence or quantifiable biomarker margins (such as Central Subfield Thickness in micrometers, Intraretinal Fluid, or Subretinal Fluid areas) required by ophthalmologists. 3) Hardware Infeasibility: Many high-parameter vision transformers and heavy 3D volumetric models require high-end GPUs, rendering them inaccessible in resource-constrained rural clinics and edge diagnostic devices. 4) Disconnection from Workflow: Existing academic models remain standalone scripts that lack integration with clinical Picture Archiving and Communication Systems (PACS), dual-curtain comparative viewing, digital calipers, automated electronic health record reporting, and multi-lingual patient counseling.

To overcome these hurdles, we developed **VisionCare**, a multi-tiered, explainable AI diagnostic workstation and clinical telemetry ecosystem. This paper documents the theoretical formulation, architectural engineering, experimental benchmarking, and clinical deployment features implemented in the VisionCare platform.

II. RELATED WORK AND CLINICAL FOUNDATIONS

Recent literature demonstrates significant progress in automated retinal diagnostic algorithms. Alyoubi et al. (2021) [1] coupled CNN512 with YOLOv3 for 5-stage DR classification and lesion localization (89.0% accuracy). Çinarer et al. (2022) [2] explored transfer learning across ResNet152, DenseNet201, and VGG19, attaining 82.7% accuracy with DenseNet201. Kotiyal & Pathak (2022) [3] developed an Apache Spark big-data pipeline with InceptionV3 achieving 95.0% binary accuracy on IDRiD. Skouta et al. (2021) [4] demonstrated 95.5% accuracy using VGG-16 with binary cross-entropy. Uppamma & Bhattacharya (2023) [5] integrated SqueezeNet with African Vulture Optimization (AVO) and blockchain storage (94.2% accuracy). Sikder et al. (2021) [6] applied gray-level intensity features with XGBoost (94.20% accuracy). Ghazal et al. (2020) [7] engineered an AlexNet CAD system on OCT images yielding 94% accuracy and 100% recall. Liu et al. (2020) [8] examined hybrid

architectures (86.34% accuracy), while Jabbar et al. (2024) [9] achieved 94% using ResNet + GoogleNet with particle swarm optimization. Moustari et al. (2024) [10] developed an Attention-Guided CNN (AG-CNN) reaching 98.48% accuracy and 0.998 AUC on APTOS 2019.

TABLE I. COMPARATIVE LITERATURE SURVEY

Model / Study	Modality	Performance	Identified Gap
CNN512+YOLO [1]	Fundus	Acc: 89.0%	High latency; no depth/fluid metrics.
InceptionV3 [3]	Fundus	Acc: 95.0%	Restricted to binary DR classes.
AlexNet CAD [7]	OCT	Acc: 94.0%	Lacks 3D macular maps & reporting.
AG-CNN [10]	Fundus	Acc: 98.48%	2D fundus only; no axial thickness.
VisionCare (Proposed)	**SD-OCT**	**Acc: 95.01% AUC: 0.9984**	**Resolves opacity; full PACS & bilingual RAG.**

III. PROPOSED SYSTEM ARCHITECTURE

A. Preprocessing and Adaptive Normalization

Input B-scans undergo contrast standardization: 1) Grayscale luminance extraction scaled to [0, 255]; 2) Contrast Limited Adaptive Histogram Equalization (CLAHE) with clip limit 2.0 and grid tile size 8x8 to enhance boundaries between the Outer Nuclear Layer (ONL) and RPE; 3) Standard model tensor normalization (3 x 224 x 224) using ImageNet channel statistics.

B. Dual-Architecture Neural Consensus Formulation

To avoid single-model bias, we pair an inverted-residual EfficientNet-B0 (MBConv with squeeze-and-excitation blocks) and a lightweight MobileNet-V3 Small backbone (depthwise-separable convolutions with Hardswish). Softmax probability distributions P_eff and P_mob across C = 11 classes are combined through linear convex weighting:

$$P_ens(y=c \mid x) = w_1\, P_eff(y=c \mid x) + w_2\, P_mob(y=c \mid x) \quad (1)$$

where empirically optimized weights w_1 = 0.70 and w_2 = 0.30 reflect relative validation feature discriminability.

C. Epistemic Uncertainty via Shannon Entropy

To identify ambiguous scans, normalized Shannon entropy is calculated across the consensus distribution:

$$H(Y \mid x) = -\,(1 / \log_2 C)\, \Sigma\, P_ens \log_2(P_ens + \varepsilon) \quad (2)$$

When H(Y | x) > 0.45 or primary diagnostic confidence is below 70%, the workstation triggers an automatic Diagnostic Triage Advisory, flagging the case for priority secondary review.

D. Attention U-Net Pathology Segmentation

Lesion morphology is delineated using an Attention U-Net. Attention Gates (AGs) filter feature activations along encoder skip connections, isolating hyporeflective intraretinal cystoid fluid cavities, subretinal fluid detachments, and sub-RPE drusenoid humps:

$$\alpha^l = \sigma_2(\, \psi^T\, (\, ReLU(\, W_x^T x^l + W_g^T g + b_g\,)\,) + b_\psi\,) \quad (3)$$

$$x_hat^l = \alpha^l \odot x^l \quad (4)$$

E. Foveal Contour Profiling and ETDRS Macular Topography

Morphological tissue analysis traces the Internal Limiting Membrane (upper boundary y_ILM(x)) and Retinal Pigment Epithelium (lower boundary y_RPE(x)). Pointwise thickness T(x) is derived via:

$$T(x) = (\, y_RPE(x) - y_ILM(x)\,) \times \delta_axial \quad (5)$$

where δ_axial = 3.9 μm/pixel is the calibrated optical axial resolution. Topographical thickness is mapped into an interactive 3D elevation surface and registered across the Early Treatment Diabetic Retinopathy Study (ETDRS) 9-zone layout: Central Fovea (1 mm), Inner Macular Ring (3 mm), and Outer Macular Ring (6 mm).

IV. EXPERIMENTAL EVALUATION

A. Dataset Partitioning and Class Distributions

The system was trained and validated on a stratified multi-class partition of the Kermany SD-OCT dataset expanded across 11 clinical diagnostic classes: CNV, DME, DRUSEN, NORMAL, AMD, CSCR, ERM, MH, RD, GLAUCOMA, and RP. Each class comprises 150 training B-scans and 31 hold-out test B-scans (Total: 1,991 B-scans).

TABLE II. KERMANY OCT BENCHMARK PARTITIONING

Class Label	Clinical Condition	Train	Test
CNV	Choroidal Neovascularization	150	31
DME	Diabetic Macular Edema	150	31
DRUSEN	Drusen Maculopathy	150	31
NORMAL	Healthy Retinal Tissue	150	31
AMD	Macular Degeneration	150	31
CSCR	Central Serous Retinopathy	150	31
ERM	Epiretinal Membrane	150	31
MH	Macular Hole	150	31
RD	Retinal Detachment	150	31
GLAUCOMA	Glaucomatous Neuropathy	150	31
RP	Retinitis Pigmentosa	150	31
Total	**11-Class Benchmark Suite**	**1,650**	**341**

B. Quantitative Diagnostic Performance

Hyperparameters: AdamW optimizer (learning rate 10^-4 with Cosine Annealing Decay down to 10^-6), batch size 16, cross-entropy loss with label smoothing (0.05). Table III summarizes classification performance across the hold-out test set (N = 341 unseen B-scans).

TABLE III. QUANTITATIVE BENCHMARK PERFORMANCE

Architecture	Accuracy	F1	ROC-AUC	Latency
MobileNet-V3	87.68%	0.8727	0.9899	68 ms
EfficientNet-B0	95.01%	0.9497	0.9984	342 ms
Consensus	**95.01%**	**0.9497**	**0.9984**	**410 ms**

C. Explainable Visual Saliency via Grad-CAM

To guarantee transparent decision rationales, Gradient-weighted Class Activation Mapping (Grad-CAM) extracts activation gradients from the final convolutional stage of EfficientNet-B0. The resulting visual heatmaps localize with high

fidelity over intraretinal fluid cysts in DME and hyperreflective subretinal membranes in CNV, providing verifiable spatial justification.

V. CLINICAL PACS TELEMETRY WORKSTATION

A. High-Performance PACS Suite with Split-Screen Curtain Slider

The clinical workstation features a Picture Archiving and Communication System (PACS) console equipped with: 1) Interactive Split-Screen Curtain Slider: Engineered via CSS clip-path polygon rendering, allowing clinicians to slide an interactive demarcation bar across the raw OCT scan and the Grad-CAM/segmentation overlay without display distortion; 2) Calibrated Digital Micrometer Caliper: An interactive canvas measurement tool calibrated to the optical axial resolution of 3.9 μm/pixel, allowing direct physical dimension measurements of foveal holes, fluid height, and sub-RPE elevation; 3) Clinical Image Controls: Real-time windowing for brightness, contrast, inversion, and sharpening.

B. Interactive 3D Macular Topography Mesh

Using the extracted ILM and RPE coordinate series, VisionCare constructs an interactive 3D topographical elevation mesh via Plotly.js, enabling clinicians to rotate, pan, and quantify macular volume and foveal contour geometry in physical coordinate space.

C. Bilingual Automated Medical Reporting (English and Tamil)

To democratize access in regional healthcare systems, an automated report generation engine compiles complete diagnostic findings in PDF format using ReportLab with embedded Unicode font rendering (Noto Sans Tamil). The reports integrate patient demographics, high-resolution original B-scans, Grad-CAM heatmaps, segmented masks, ETDRS thickness statistics, and localized clinical lifestyle and therapeutic advice.

D. Diagnostic Thought-Chain Conversational RAG Assistant

An integrated conversational assistant powered by Retrieval-Augmented Generation (RAG) assists ophthalmologists with differential diagnosis. The engine references indexed clinical trial literature and ETDRS treatment protocols. Each response is preceded by a structured reasoning trace detailing internal inference steps. Hands-free voice input and speech synthesis enable sterile, hands-free operation in clinical examination rooms.

VI. CONCLUSION AND FUTURE SCOPE

In this work, we introduced VisionCare, an explainable, multi-class retinal OCT diagnostic and clinical PACS telemetry ecosystem. By combining EfficientNet-B0 and MobileNet-V3 into a weighted consensus ensemble (95.01% accuracy, 0.9984 ROC-AUC), quantifying epistemic uncertainty through Shannon entropy, and integrating an Attention U-Net for spatial lesion extraction, the system delivers high accuracy with sub-450 ms CPU inference latency. Future research will extend the consensus framework to 3D volumetric OCT C-scans and multimodal data pairing.

ACKNOWLEDGMENT

The authors express their gratitude to the Department of Computer Science & Engineering and Department of Biomedical Engineering for computational resources and institutional support throughout the development of the VisionCare project.

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